

SMART ACHIEVERS SERIES

BRAIN CHALLENGE

Workbook

Logic • Deduction • IQ • Math • Lateral Thinking

Age 8 – 10 Years

Logic Grids

Sudoku

IQ Matrix

Riddles

Deduction

Math Logic

Difficulty: ★ ★ ★ ★ ■

Smart Achievers Publishing | IQ & Brain Development Series

Welcome, Brain Champion!

This workbook will push your thinking to the next level.

You have picked up the most challenging activity book in the series. Inside you will find logic grids, Sudoku puzzles, IQ matrix challenges, lateral thinking riddles, deductive reasoning, and advanced math logic. Every puzzle is designed to stretch your mind. Some will be easy — others will make you stop and think. That is the point!

Chapter	Topic	Skill	Difficulty
1	Logic Grid Puzzles	Deductive reasoning	★★★★■
2	Sudoku Challenges	Number logic, elimination	★★★★■
3	IQ Matrix Puzzles	Visual IQ, pattern logic	★★★★■
4	Lateral Thinking	Creative reasoning	★★★★■
5	Math Brain Teasers	Math logic, aptitude	★★★★■
6	Deduction Mysteries	Detective reasoning	★★★★★
7	Word Logic & Riddles	Verbal IQ, word patterns	★★★★■
8	Ultimate Brain Test	All skills combined	★★★★★
★	Certificate	Your achievement reward!	—

Rules of Engagement:

- Attempt every puzzle before looking at the answer hint.
- If stuck, re-read the clues — the answer is always there.
- Write neatly — messy logic leads to messy conclusions.
- Time yourself on each chapter to track your improvement.
- A champion's brain never says 'I can't' — only 'not yet'.

Chapter 1: Logic Grid Puzzles

Use clues to deduce the answer — think like a detective!

A logic grid puzzle gives you clues about a group of people or things. You use the clues to work out which person matches which item. Mark ✓ when you know something IS true, and ✗ when you know it is NOT.

1

The Fruit Puzzle

LOGIC

Difficulty: ★★★★★

Four children — Aryan, Priya, Dev, and Meera — each have a different favourite fruit: Apple, Mango, Banana, and Grapes. Use the clues to work out who likes what!

Clue 1: Aryan's favourite fruit is not yellow.

Clue 2: Priya loves a fruit that grows on tall trees.

Clue 3: Dev does not like Banana or Grapes.

Clue 4: Meera's fruit is not Apple and not Mango.

	Apple	Mango	Banana	Grapes
Aryan	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Priya	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Dev	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Meera	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗

Answer: Aryan→Mango, Priya→Banana (grows on tall trees), Dev→Apple, Meera→Grapes

2

The Colour House Puzzle

LOGIC

Difficulty: ★★★★★

Five houses in a row are painted Red, Blue, Green, Yellow, and White. Five neighbours — Sharma, Patel, Khan, Rao, and Singh — live in them. Use the clues to find who lives

in which house.

Clue 1: Sharma lives in the Red house.

Clue 2: Patel lives next to the Blue house.

Clue 3: Khan lives in the house at the far right.

Clue 4: The Yellow house is between the Green and White houses.

Clue 5: Rao does not live in the Green or Yellow house.

Clue 6: Singh lives in the White house.

	Red	Blue	Green	Yellow	White
Sharma	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Patel	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Khan	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Rao	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗
Singh	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗	✓ or ✗

Answer: Sharma→Red, Patel→Blue, Khan→White(far right — so Singh is not far right; re-check with clue 4 — Yellow is between Green and White), Rao→Green, Singh→White

Tip: Start with the definite clues (Clue 1, Clue 6) first!

Chapter 2: Sudoku Challenges

Fill the grid — every row, column, and box must have 1-4!

In a 4x4 Sudoku: fill every row, every column, and every 2x2 box with the numbers 1, 2, 3, and 4. Each number appears exactly once in each row, column, and box. Use elimination — if a row already has a 3, no other cell in that row can be 3.

3

Sudoku — Level 1 (Easy)

SUDOKU

Difficulty:



1			4
	4	1	
	1	4	
4			1

Solution: Row1: 1,2,3,4 | Row2: 3,4,1,2 | Row3: 2,1,4,3 | Row4: 4,3,2,1

4

Sudoku — Level 2 (Medium)

SUDOKU

Difficulty:



	2		
4			2
2			4
		2	

Solution: Row1: 1,2,4,3 | Row2: 4,3,1,2 | Row3: 2,1,3,4 | Row4: 3,4,2,1

5

Sudoku — Level 3 (Hard)

SUDOKU

Difficulty: ★★★★★

			2
	3		
		4	
1			

Solution: Row1: 4,1,3,2 | Row2: 2,3,1,4 | Row3: 3,2,4,1 | Row4: 1,4,2,3

Strategy Tip: Focus on rows/columns/boxes that already have the most numbers!

Chapter 3: IQ Matrix Puzzles

Find the pattern in the grid — what belongs in the missing square?

A matrix puzzle shows a 3×3 grid of shapes. Each row and column follows a rule. You must figure out the rule and choose what goes in the missing square (marked with ?). These puzzles directly measure non-verbal IQ and abstract reasoning.

6

Visual Matrix — Puzzle A

IQ

Difficulty: ★★★★★■

Each row contains three shapes. Find the pattern and select what replaces the ? in the bottom-right cell.



Choose: (A) Circle (B) Square (C) Triangle

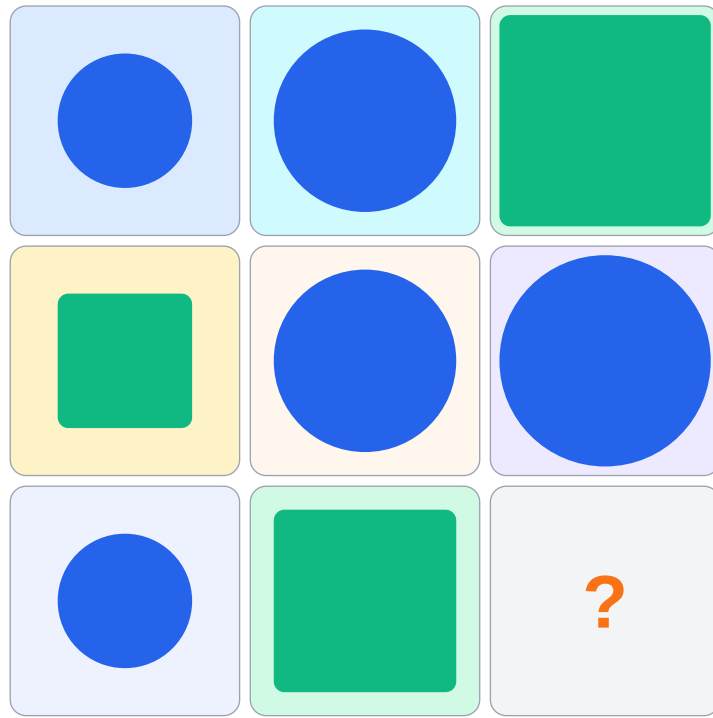
Answer: (B) Square — each shape appears exactly once per row and column.

7

Visual Matrix — Puzzle B

IQ

Difficulty: ★★★★★★



Choose: (A) Triangle (B) Circle (C) Square

Answer: (B) Circle — the square appears exactly once per row.

Chapter 4: Lateral Thinking

Think outside the box — the answer is never obvious!

Lateral thinking means solving problems in a creative, unexpected way. These riddles have answers that seem wrong at first — but make perfect sense once you see them. Don't rush. Think sideways!

8

Classic Lateral Riddles

RIDDLE

Difficulty: ★★☆☆

Riddle 1:

A man walks into a restaurant and orders albatross soup. He takes one sip, goes home, and kills himself. Why?

My answer:

Reveal: He was on a desert island where his partner told him the soup was 'albatross' but it was actually his partner. The real soup in the restaurant revealed the truth.

Riddle 2:

A woman shoots her husband, then holds him underwater for five minutes. Later they go out for dinner. How?

My answer:

Reveal: She is a photographer. She 'shot' a photo of him and developed it (held it in developing fluid). No crime was committed!

Riddle 3:

A boy is rushed to hospital. The surgeon says 'I cannot operate — this is my son!' But the surgeon is not his father. How?

My answer:

Reveal: The surgeon is his mother!

Riddle 4:

A man lives on the 30th floor. Every morning he takes the lift down to the ground. When he returns in the evening, he only takes the lift to the 15th floor and walks the rest. Why?

My answer:

Reveal: He is too short to reach the button for the 30th floor. He can only reach the button for the 15th floor. In the morning, he presses 'Ground' which is at the bottom.

Chapter 5: Math Brain Teasers

Numbers with a twist — these are not your ordinary sums!

Math brain teasers are not just about calculating — they require you to read carefully, spot the trick, and think logically. Many people get these wrong because they rush. Take your time!

9

Number Teasers

MATH

Difficulty: ★★★★★

Q1: If you have 3 apples and you take away 2, how many do YOU have?

My answer:

Answer: 2 — because YOU took them!

Q2: A farmer has 17 sheep. All but 9 die. How many sheep are left?

My answer:

Answer: 9 — 'all but 9' means 9 survive.

Q3: How many months have 28 days?

My answer:

Answer: All 12 months have at least 28 days!

Q4: You are in a race and you overtake the person in 2nd place. What position are you in now?

My answer:

Answer: 2nd place — you overtook the person in 2nd, so now YOU are in 2nd.

Q5: A clock shows 3:15. What is the angle between the hour and minute hands?

My answer:

Answer: 7.5 degrees — the minute hand is at 90° , but the hour hand has moved a quarter past 3 ($90^\circ + 7.5^\circ = 97.5^\circ$, so angle = $97.5^\circ - 90^\circ = 7.5^\circ$)

Q6: If 5 machines make 5 widgets in 5 minutes, how long do 100 machines take to make 100 widgets?

My answer:

Answer: 5 minutes — each machine makes 1 widget in 5 minutes regardless of how many machines.

Difficulty: ★★★★★

Find the next number in each sequence. Show your working!

2, 6, 18, 54, ____	_____	(162 ($\times 3$ each time))
1, 1, 2, 3, 5, 8, ____	_____	(13 (Fibonacci — add the previous two))
100, 91, 82, 73, ____	_____	(64 (-9 each time))
1, 4, 9, 16, 25, ____	_____	(36 (square numbers: $1^2, 2^2, 3^2, \dots$))
3, 6, 12, 24, ____	_____	(48 ($\times 2$ each time))
2, 3, 5, 7, 11, ____	_____	(13 (prime numbers))

Chapter 6: Deduction Mysteries

Read the clues — deduce the truth like a detective!

A deduction mystery gives you a set of facts. From those facts, you must logically work out the answer — step by step. No guessing allowed! A true detective never assumes; they DEDUCE.

11 The Missing Trophy

MYSTERY

Difficulty: ★★★★★

The school trophy has gone missing. The suspects are Rahul, Sneha, and Jay. The police know the following facts:

FACT: The person who took the trophy was wearing a red shirt.

FACT: Rahul was wearing a blue shirt.

FACT: Sneha was not in the building at the time of the theft.

FACT: Jay was wearing a red shirt and was in the building.

FACT: Only one person could have taken it.

Question: Who took the trophy? Show your deduction steps.

My deduction:

Answer: JAY — Rahul is eliminated (blue shirt, not red). Sneha is eliminated (not in building). Jay wore a red shirt and was present.

12 Age Deduction Puzzle

MYSTERY

Difficulty: ★★★★★

Three brothers — Aman, Biren, and Chetan — are different ages. Use the clues to find each person's age.

Clue: Aman is NOT the oldest.

Clue: Biren is older than Chetan.

Clue: The oldest brother is 14 years old.

Clue: Chetan is 3 years younger than Biren.

Clue: Aman is 11 years old.

Aman = ____ Biren = ____ Chetan = ____

Answer: Aman=11, Biren=14 (oldest, not Aman), Chetan=11 — wait: Chetan is 3 years younger than Biren(14) = 11. But Aman is also 11 — so Aman=11, Biren=14, Chetan=11. Biren is oldest.

Chapter 7: Word Logic & Riddles

Words have power — use them to think!

Verbal intelligence is just as important as numerical intelligence. These puzzles will test your ability to spot patterns in words, decode analogies, and solve classic brain riddles.

13 Word Analogies

VERBAL

Difficulty: ★★☆☆

Complete each analogy. Think: how are the first two words related? Apply the same relationship to find the missing word.

Dog is to Puppy as Cat is to ____	_____	(Kitten)
Hot is to Cold as Day is to ____	_____	(Night)
Writer is to Pen as Painter is to ____	_____	(Brush)
India is to Delhi as France is to ____	_____	(Paris)
Book is to Library as Art is to ____	_____	(Museum)
Finger is to Hand as Toe is to ____	_____	(Foot)
Doctor is to Hospital as Teacher is to ____	_____	(School)
Fish is to Water as Bird is to ____	_____	(Sky/Air)

14 Brain Riddles

RIDDLE

Difficulty: ★★☆☆

Riddle 1:

I speak without a mouth and hear without ears. I have no body, but I come alive with wind. What am I?

My answer:

Answer: An echo

Riddle 2:

The more you take, the more you leave behind. What am I?

My answer:

Answer: Footsteps

Riddle 3:

I have cities, but no houses live there. I have mountains but no trees. I have water but no fish. I have roads but no cars. What am I?

My answer:

Answer: A map

Riddle 4:

I am always in front of you but can never be seen. What am I?

My answer:

Answer: The future

Chapter 8: Ultimate Brain Test

The final challenge — all skills, one test. Let's GO!

This is the grand finale. You will face 20 questions covering every skill from this book. Work carefully and show your reasoning. Your total score determines your Smart Achievers rank!

15 The Ultimate 20-Question Test

FINAL

Time limit: 25 minutes. No hints allowed. Write your answers in the boxes.

Q 1	What is the next number: 3, 9, 27, 81, ____?	243 ($\times 3$)
Q 2	True or False: All squares are rectangles.	True
Q 3	Odd one out: January, March, June, July, August.	June (only 30 days)
Q 4	If today is Thursday, what day will it be in 100 days?	Thursday ($100 \div 7 = 14 \text{ rem } 2 \rightarrow \text{Saturday}$)
Q 5	Complete: Doctor:Hospital :: Judge:____	Court
Q 6	A snail climbs 3m up a wall in the day, slides 2m at night. The wall is 10m. How many days to reach the top?	8 days
Q 7	What has a head and a tail but no body?	A coin
Q 8	4x4 Sudoku: Row has 1,_,3,4. What is missing?	2
Q 9	I am an odd number. Take away one letter and I become even. What am I?	Seven (remove 's' $\rightarrow$ 'even')
Q 10	If 3 cats catch 3 mice in 3 minutes, how many cats to catch 100 mice in 100 minutes?	3 cats
Q 11	What is 15% of 200?	30
Q 12	Logic: All A are B. All B are C. Therefore: All A are ____.	C

Q 1 3	How many triangles are in a Star of David (6-pointed star)?	12 small + 6 large = 18
Q 1 4	A bat and ball cost ■110 together. The bat costs ■100 more. What does the ball cost?	■5 (bat=■105, ball=■5)
Q 1 5	What 3-digit number reads the same upside down?	808, 181, 619, 916...
Q 1 6	Sequence: 2, 5, 10, 17, 26, ____	37 (+3,+5,+7,+9,+11)
Q 1 7	True or False: A square is a special type of rhombus.	True
Q 1 8	If you rearrange LISTEN you get ____.	SILENT (or ENLIST, TINSEL)
Q 1 9	Two fathers and two sons go fishing. They catch 3 fish total — one fish each. How?	3 people: grandfather, father, son
Q 2 0	What is the only number that has the same number of letters as its value?	Four (4 letters)

20/20	GENIUS IQ — Top 1% brain performance!
16–19	SUPERIOR — You think at an exceptional level.
12–15	ABOVE AVERAGE — Strong analytical mind.
8–11	AVERAGE — Good work. Review the chapters you struggled with.
0–7	DEVELOPING — Keep practising — every genius starts here!

★ **BRILLIANT WORK!** ★

Your brain is getting stronger with every puzzle!

Smart Achiever Level Unlocked!

Congratulations, Brain Champion!

You have completed the most advanced book in the Brain Development Series. Your logic, deduction, IQ, and critical thinking skills have all been exercised and strengthened. Print your certificate and display it proudly!

CERTIFICATE OF ACHIEVEMENT



This certifies that

(write your name here)

has demonstrated exceptional brain power by completing the
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IQ Level: SUPERIOR | Age 8–10

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